

Thin-film lithium niobate for quantum photonics: perspectives and challenges

Yunhong Ding⁽¹⁾, Innokentiy Zhdanov⁽¹⁾, and Leif K. Oxenløwe⁽¹⁾

⁽¹⁾ Technical University of Denmark, Department of Electrical and Photonics Engineering, 2800 Kongens Lyngby, Denmark (yudin@dtu.dk)

Abstract *Thin-film lithium niobate (TFLN) photonics has recently emerged as an appealing platform for quantum photonics due to its low-loss characteristics, large electro-optic and nonlinear coefficients, and broad transparency window. Here, we discuss the latest developments, promising directions, and challenges for future TFLN quantum photonics research. ©2026 The Author(s)*

1. Introduction

Quantum photonics is very promising to lead to a step change in human capabilities in various applications. Quantum photonic computers may outperform classical supercomputers for certain computational tasks [1]. Quantum photonic network would lead to secure communications [2] and interconnecting multiple quantum devices [3]. Quantum photonic simulators with efficient quantum photonic circuits are able to effectively simulate problems e.g. quantum chemistry [4,5], spin disordered systems [6]. Quantum metrology is also able to facilitate multiparameter estimation with enhanced sensitivity [7].

Scaling the quantum systems up is essential for enhancing the capabilities of quantum photonic technologies, which is however challenged by the sensitivity of these systems to environmental factors such as optical losses, phase errors, limited light-matter interaction etc. Integrated quantum photonics offers a promising path forward, providing miniaturized scalable platforms with reduced size and power consumption compared with conventional free-space optics systems and, in particular, offers phase-stable optical paths, which greatly eliminate the environmental disruptions.

Besides aforementioned sensitivity, various quantum technologies require extreme performance from the integrated quantum photonics. Quantum networks which realize transmission of quantum states between distant parties, are highly relying on low-loss modulators operating at high speed (tens of GHz clock rate), low-loss fiber-to-chip couplers (<1dB) and high-efficiency detectors (>90%). Quantum metrology, which is able to surpass classical measurement limits, is highly dependent on the thermal stability and nonlinearity, which are critical for enhancing measurement precision.

We are currently in the era of noisy intermediate-scale quantum (NISQ) technologies [8] where the goal is to surpass classical supercomputers tackling specific problems such as quantum simulations [9], optimization problems [10],

and particular tasks that challenge classical algorithms [11], utilizing experimental capabilities that are currently within reach. Expanding quantum computing power of NISQ systems requires integrating thousands of optical components on-chip with high yield, precision and low optical loss. Furthermore, large-scale of the NISQ technologies rely on near-deterministic, predictable generation of single photons or multi-photon entanglement quantum states. A promising means is based on multiplexing technology to convert heralded single-photon inputs or heralded multi-photon entanglement states to on-demand states by detecting the heralding photon and reallocating the photons to the predetermined target output mode through fast active feedforwarding [12]. Achieving this requires an ultra-low loss photon storage loop and fast switch. The ultimate goal for quantum information processing is to achieve universal fault-tolerant photonic quantum computing.

The emerging thin-film lithium niobate (TFLN)-based platforms with loss metrics comparable with the best of other integrated photonics platforms (e.g. silicon nitride (SiN)), offer significant advantages across a broad spectral range. TFLN holds strong second- and third-order nonlinearity enabling efficient nonlinear processes for photon pair generation. The cryo-compatible low-loss and ultrafast modulation through the efficient Pockels effect makes it superior to Si and SiN based platforms. In addition, LN is extremely promising for on-chip quantum memories [13] thanks to the excellent crystalline host for rare-earth ions [14]. The other missing components e.g. single-photon detectors (SPDs) can be potentially heterogeneously integrated.

In this paper, we provide a comprehensive discussion on the state-of-the-art TFLN quantum photonics and analyze its perspectives and potential future challenges from component level to the applications.

2. TFLN photonics platform

Since the 1960s, bulk LN waveguide fabricated by proton exchange (PE) processes [15,16] or metal in-diffusion [17] with weak light confinement have been studied for electro-optic modulation and nonlinear optical processes. This weak mode confinement led to reduced nonlinear efficiency, and large bending radii and significant footprints [18] which significantly limits the application in quantum photonics. The emergence of TFLN wafers, in the 2000s, has positioned TFLN as one of the most promising candidates for integrated photonic platforms. The state-of-the-art fabrication based on Ar plasma etching lead to a record propagation loss $<1.3\text{dB/m}$ [19] which is comparable with the state-of-the-art silicon nitride photonics platform.

2.1 Interfacing with fiber

Like all integrated photonics, fiber-to-chip coupling is essential for TFLN photonics devices to interface with fibers. Grating couplers are convenient options able to couple with standard single mode fibers. Due to the power leakage loss to the substrate, the coupling efficiency is typically limited, and the 1-dB coupling bandwidth is limited to tens of nm. To achieve higher coupling efficiency, metal reflectors are typically applied, leading to a low coupling loss of 0.89 dB [20]. Compared with grating couplers, edge couplers provides much broader coupling bandwidth (typically 1dB bandwidth $>100\text{nm}$), and coupling loss $<1\text{dB}$ was widely reported. The record low coupling loss of 0.29dB and 0.24 dB for the TE and TM modes to the high numerical-aperture fiber (HNAF) is recently reported [21].

2.2 TFLN switches and modulators

The efficient Pockels effect of LN allows pure phase electro-optic (E/O) modulation, which is desirable for quantum photonics, e.g. high-fidelity demultiplexing of quantum dot single photon sources (SPSs) [22], multiplexing of nonlinear sources [23], and universal quantum computing [24]. Significant progress on the TFLN modulators was achieved during the past 5 years. CMOS-level driving voltages being reported [25]. Other major breakthroughs include a 2 cm long EO modulator with a V_π of only 1.3 V [26], high bandwidths of 67 GHz and driving voltages of 3.1 V [27], and recent dual-polarization IQ modulators [28] which is very useful for continuous variable (CV) quantum key distributions (QKD).

2.3 Heterogenous integration with quantum emitters on TFLN

As the TFLN platform lacks efficient deterministic quantum emitter, a natural idea is to heterogeneously integrate quantum emitters of other material platforms onto the TFLN platform. The first heterogenous integration of InAs quantum dot on

the TFLN platform was demonstrated [29] in 2018 using the “pick-and-place” approach [30]. The total collection efficiency of 9% was reported, considering a QD emission efficiency (85%) and grating coupler efficiency of (26%). GaAs QD emitter was also successfully integrated on the TFLN platform, with 42% overall source efficiency and almost lossless transition of the single photon to the TFLN chip [31].

2.4 Heterogeneous integration with detectors

Super-conductive nanowire single photon detectors (SNSPDs) are most attractive to be heterogeneously integrated on the TFLN platform, thanks to the high detection efficiency and low jitter. Key progress includes the integration of a NbN nanowire on a TFLN waveguide with 46% on-chip detection efficiency and 32 ps timing jitter [32], NbTiN SNSPD on the TFLN waveguide with 27% on-chip detection efficiency and 17 ps jitter [33], and MoSi nanowire on TFLN with 50% on-chip detection efficiency and 82 ps jitter at 1550 nm [34].

3. Quantum photonic applications

3.1 Nonlinear quantum frequency conversion

Quantum frequency conversion (QFC) is typically applied to link different technologies optimized at different wavelengths, e.g. down-converting single photon sources at 900nm [35] to 1550nm [36] being compatible with low-loss fiber transmission, and up-converting long wavelength photons to near visible range for using Si detectors. The most attractive feature for LN is that the crystalline can be periodically poled to achieve quasi-phase matching. Such so-called PPLN waveguide is very efficient to achieve high-efficiency QFC. Significant progress includes 73% up-conversion efficiency from 1550nm to 863nm [37] and 95% down-conversion efficiency from 780nm to 1541nm [38].

3.2 Photon pair generation

Heralded photon pair source (PPS) can be effectively achieved by TFLN photonics platform via nonlinear spontaneous parametric down conversion (SPDC). Key metrics include pair generation rate (PGR), coincidence-to-accidental ratio (CAR), second-order autocorrelation at zero delay of a heralded single-photon $g^{(2)}(0)$, and visibility of two photon interference. Typically, there is a trade-off between them. Photon-pair generation has investigated with the use of both TFLN devices and Ti-diffused waveguides. TFLN devices are particularly attractive thanks to the tighter light confinement leading to stronger light-matter interaction and more efficient photon pair generation. Both periodic poling and modal phase matching (MPM) were investigated. MPM TFLN sources achieved a record high CAR of 58298 ± 1297 with $g^{(2)}(0) < 0.2$ and generation rate

of 41 GHz/mW [39], while high CAR of 67224 ± 714 with $g_h^{(2)}(0)$ of 0.022 ± 0.0004 and PGR of 0.045 GHz/mW was achieved by periodically poled TFLN (PPTFLN) waveguide source [40], and CAR of 14682 ± 4427 with $g_h^{(2)}(0)$ of 0.022 ± 0.0004 and PGR of 2.6 ± 0.1 GHz/mW was achieved by PPTFLN ring resonator source [41].

The nonlinear photon PPSs are suffering from the nature of probabilistic generation. A promising approach towards a near deterministic generation of nonlinear SPSs is based on active multiplexing which requires low photon storage and fast and high-speed switches. For this perspective, TFLN photonics is an ideal contender platform. Single photon storage [42] and temporal multiplexing [12] have already been investigated by integrated TFLN photonic devices, as presented in Fig. 1, showing an effective boost of generation probability. For further improvement towards the goal of near deterministic generation, hybrid spatial and temporal multiplexed photon source is envisioned [43].

3.3 Squeezed light source

Squeezed light, where one of the quadratures achieves noise distribution lower than the shot noise level [44], is an essential resource for CV quantum communications, quantum computing,

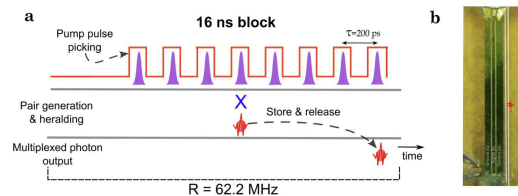


Fig. 1: a. Illustration of the principle of temporal multiplexing with the values employed in [12]. A photon pair source pumped with a period $\tau = 200$ ps generates photons in N different time bins probabilistically. A storage delays signal photons for an arbitrary integer multiple of τ , allowing any initial time-bin state of a heralded photon to be converted to a fixed output time-bin. b. Optical microscope image of a single TFLN storage loop with connected high-speed radio frequency and heater probes.

and quantum metrology. Squeezed light has been investigated by bulk LN photonics, the strong light confinement and light-matter interaction make PPTFLN platform particularly attractive. So far, a moderate squeezing of -0.56 ± 0.09 dB (inferred on-chip -2.6 dB) was reported [45], primarily limited by the coupling loss and nonuniformity of the poling. An on-chip squeezing of -4.9 dB by TFLN waveguide was also recently reported [46].

3.4 Quantum metrology

Squeezed states of light are particularly attractive in quantum metrology as they can be bright and enhance sensitivity by reducing the noise level below the shot noise in their squeezed quadrature [47]. Recent advancements have reported

the integration of squeezed light sources into photonics circuits on the same TFLN chip, achieving nearly complete phase sensor functionality with quantum-enhanced sensitivity [48].

3.5 Quantum computing

The investigations introduced above on components and quantum light of sources lay a clear path for TFLN photonics towards quantum computing, which requires high quality sources, low-loss feature, and high-speed and high-fidelity manipulation of quantum state of light. Recently, integration of single photon demultiplexing circuit with a TFLN reconfigurable multimode unitary interferometer in both Reck (triangle) scheme [49] and the Clements (rectangular) scheme [50] are reported, an important step towards reconfigurable quantum computing. Furthermore, the squeezed light source, high-speed and low loss devices delivered from the TFLN platform would highlight the potential for CV measurement-based quantum computation (MBQC). The Loop-based CV MBQC architectures with integrated switches [51] could greatly benefit from TFLN photonics. Moreover, the TFLN photonic circuits would find promising applications for programmable local oscillators for on-chip phase-dependent homodyne detection [52].

Challenges and outlook

TFLN photonics has emerged as a powerful and versatile platform for quantum photonics applications, thanks to its unique features of low-loss, ultra-fast electro-optic switching capabilities, and high nonlinear coefficients. The applications cover a wide spectrum ranging from enhanced quantum metrology, active feedforwarding for multiplexing, high-secret-key-rate QKD, and efficient quantum state sources for universal quantum computing and quantum internet. However, for further scalability, a few key challenges of TFLN quantum photonics remains to be tackled. First of all, despite the breakthrough of record low propagation loss of 1.3dB/m, there is still room to improve to the theoretical limit of ~ 0.1 dB/m [53], which would lead to much more efficient multiplexing technology and near-deterministic nonlinear SPS. Secondly, low-loss and high-extinction ratio (>100 dB) pump filters are still missing on the TFLN platform, which is indispensable for realizing on-chip SNSPDs. Thirdly, fiber-to-chip coupling still needs to be further improved to the level achieved for the state-of-the-art SiN to be 0.052 ± 0.012 dB coupling loss with fiber [54]. A heterogeneous SiN-TFLN platform would be a compelling active research direction. Finally, the photon number resolving (PNR) detectors are still to be developed on the TFLN platform, which is required for multiplexing and NISQ devices, e.g. Gaussian boson sampler (GBS).

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